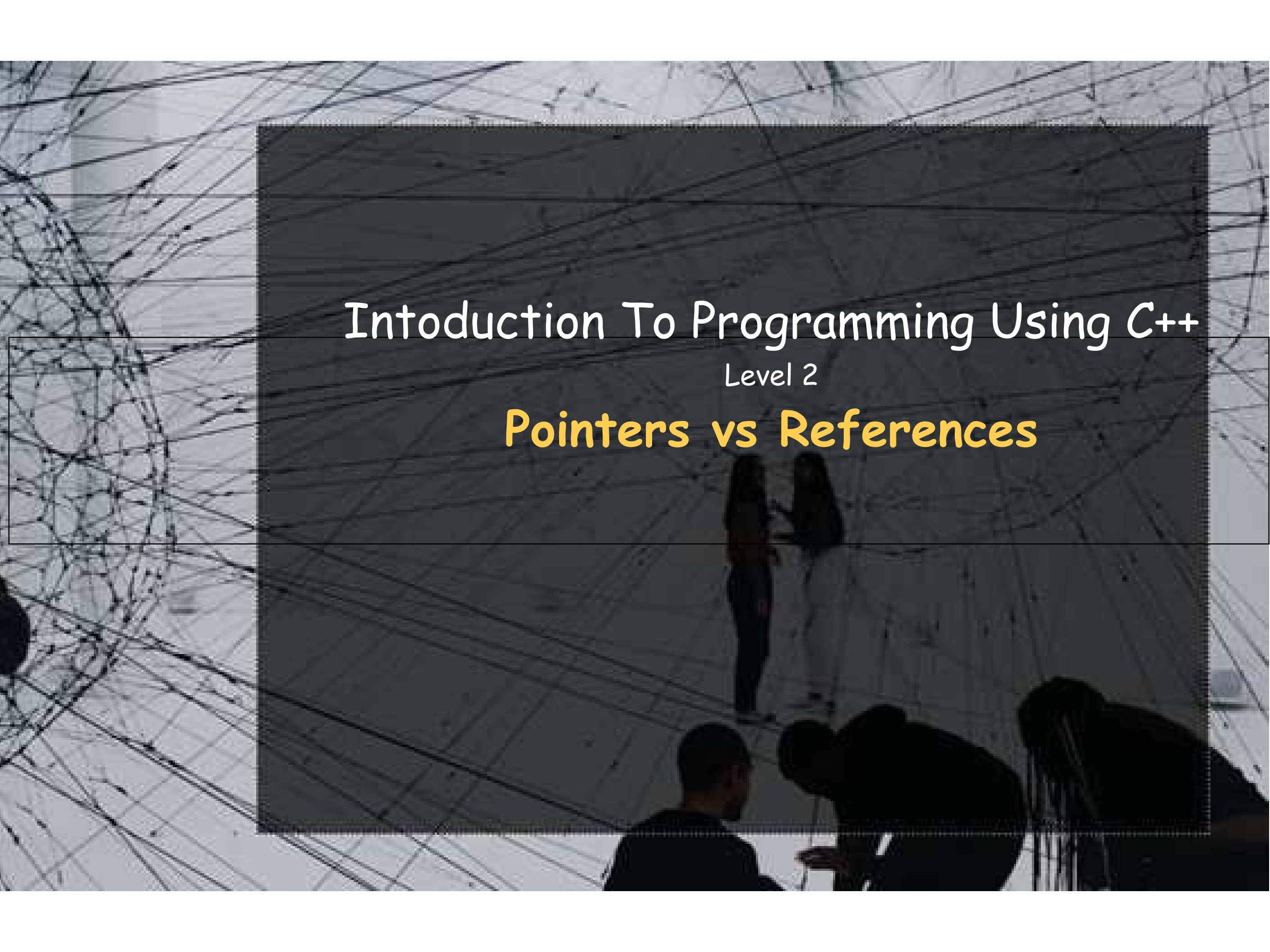


Introduction To Programming Using C++ Level 2

Pointers vs References



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Pointers vs References

Definition

Reference

-  A Reference is another name (Alias) for an existing variable.

-  A reference refers directly to the same variable and does not create a separate variable.

Pointer

-  A Pointer is a separate variable that stores the memory address of another variable.

-  The pointer has its own memory address and stores the address of the variable it points to.

Explanation

Reference

-  A reference is created using the & symbol.
-  Once a reference is initialized, it remains bound to the same variable.
-  A reference is used like the original variable.
-  You do not need to use * to access the value through a reference.

Pointer

-  A pointer is declared using the * symbol.
-  A pointer stores the address of another variable.
-  The pointer can be changed to point to another compatible variable.
-  The * operator is used to dereference the pointer and access the value.

Procedures

1. Create an original variable.
2. Create a reference to the variable.
3. Create a pointer to the same variable.
4. Compare the addresses of the original variable and the reference.
5. Print the pointer to see the stored address.
6. Use `*p` to access the value through the pointer.
7. Create another variable.
8. Change the pointer to point to the new variable.
9. Observe that the reference remains bound to the original variable.

💡 Example

◆ Reference and Pointer

```
int main()
{
    int a = 10;
    int& x = a;    // Reference
    int* p = &a;   // Pointer
    cout << "&a = " << &a << endl;
    cout << "&x = " << &x << endl;
    cout << "p = " << p << endl;
    cout << "*p = " << *p << endl;
    int b = 20;
    p = &b;
    cout << "After p = &b:" << endl;
    cout << "*p = " << *p << endl;
    cout << "a = " << a << endl;
    cout << "x = " << x << endl;
    return 0;
}
```

➡ Result

&a and &x have the same address.
p contains the address of a.

Note

Reference

 References are commonly useful for function parameters because they allow a function to work with the original variable without making a copy.

Pointer

 Pointers are useful when we need to change what is being pointed to, work with dynamically allocated objects, or represent an optional absence of an object using `nullptr`.

⊘ Warning

- ◆ Do not think that a reference and a pointer are exactly the same thing.
- ◆ A reference is an alias for an existing object, while a pointer is a separate object that stores an address.
- ◆ Be careful with pointers because dereferencing an invalid or null pointer causes undefined behavior.

Comparison

Reference

- ✓ An alias for an existing variable.
- ✓ Refers directly to the same variable.
- ✓ Must be initialized when declared.
- ✓ Cannot be changed to refer to another variable.
- ✓ Used like a normal variable.
- ✓ Does not use * for normal access.

Pointer

- ✓ A separate variable that stores an address.
- ✓ Has its own memory address.
- ✓ Can be changed to point to another compatible variable.
- ✓ Uses * to dereference and access the pointed-to value.
- ✓ Can be initialized with nullptr when it should point to nothing.

✦✦ Summary

- ✓ Reference → Another name (Alias) for an existing variable.
- ✓ Pointer → A separate variable that stores the address of another variable.
- ✓ Reference → Cannot be redirected to another variable after initialization.
- ✓ Pointer → Can be redirected to another compatible variable.
- ✓ Reference → Used like the original variable.
- ✓ Pointer → Use * to dereference and access the pointed-to value.
- ✓ Reference → Directly refers to the same variable.
- ✓ Pointer → Stores the address of the variable it points to.

Thank you 😊